Fresnel and polarization compensation in a stigmatic diopter

Model behind notebook 04

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Abstract

A Cartesian oval Σ images an axial point without aberration, yet local Fresnel transmission, the tilt of the p field and the ray-tube amplitude variation still distort the vector focus. We derive how a transmitted field on Σ maps to the angular spectrum of the image space, why that map is the stationary-phase (Debye) limit of an exact surface-to-spectrum integral, and how dividing the entrance field by the resulting p and s transfer weights makes the transverse focal field exactly the Airy pattern. No aplanatic objective, sine condition or Richards–Wolf apodization is assumed.

1 Geometry

The object point $P_0 = (0, 0, z_0)$ lies in a medium of index n_o, and its image $F = (0, 0, z_i)$ in a medium of index n_i. The surface Σ is the vertex-connected Cartesian oval (GOTS form) with the vertex at the origin:

$$n_o\,|Q - P_0| + n_i\,|F - Q| = n_o|z_0| + n_i z_i \qquad \forall\, Q \in \Sigma. \tag{1}$$

Every ray from P_0 therefore refracts towards F, and every ray reaches F with the same optical path. Notebook 04 uses

Quantity	Value
Object conjugate z_0	-10 mm (air, $n_o = 1$)
Image conjugate z_i	6 mm (fused silica, $n_i = 1.46$, inferred)
Wavelength λ_0	532 nm
Image-space NA $= n_i \sin\theta_{\max}$	0.84096
Clear semi-aperture a (surface radius)	2.29106 mm
Sag at the rim	2.7486 mm
Largest incidence angle on Σ	88.57°

Points of Σ are labelled by their cylindrical radius r and azimuth ϕ; $\boldsymbol{e}_\rho, \boldsymbol{e}_\phi, \boldsymbol{e}_z$ is the local cylindrical basis. For each $Q(r, \phi)$ let $\boldsymbol{n}$ be the unit normal, $\boldsymbol{d}_o = (Q - P_0)/|Q - P_0|$ the incident direction, $\boldsymbol{d}_i$ the Snell transmitted direction, $L_i = |F - Q|$ and $\cos\theta = \boldsymbol{d}_i \cdot \boldsymbol{e}_z$. Stigmatism means $\boldsymbol{d}_i = (F - Q)/L_i$; the notebook verifies this to 6×10^{-16} and the constancy of (1) to $2 \times 10^{-11}\lambda_0$.

2 Entrance field and local transmission

Entrance field. With $\tan a_0 = r/(z_Q - z_0)$ the tilt of the incident ray, an x-polarized field made transverse to $\boldsymbol{d}_o$, with independent p and s amplitudes $a_p(r)$, $a_s(r)$, is

$$\boldsymbol{E}_0 = a_p \cos\phi\,(\boldsymbol{e}_\rho - \tan a_0\,\boldsymbol{e}_z) - a_s \sin\phi\,\boldsymbol{e}_\phi. \tag{2}$$

Indeed $\boldsymbol{d}_o = \sin a_0\,\boldsymbol{e}_\rho + \cos a_0\,\boldsymbol{e}_z$ gives $\boldsymbol{E}_0 \cdot \boldsymbol{d}_o = 0$. For $a_p = a_s = 1$ the transverse components are constant, $E_{0x} = 1$, $E_{0y} = 0$: this is the *uncompensated* illumination.

Local Fresnel transmission. With $\boldsymbol{s} = \boldsymbol{d}_o \times \boldsymbol{n}/|\cdot|$, $\boldsymbol{p}_o = \boldsymbol{s} \times \boldsymbol{d}_o$ and $\boldsymbol{p}_i = \boldsymbol{s} \times \boldsymbol{d}_i$,

$$\boldsymbol{E}_t(Q) = \mathsf{T}\,\boldsymbol{E}_0, \qquad \mathsf{T} = t_p\,\boldsymbol{p}_i \otimes \boldsymbol{p}_o + t_s\,\boldsymbol{s} \otimes \boldsymbol{s}, \tag{3}$$

with the electric Fresnel coefficients $t_s = 2n_o c_o/(n_o c_o + n_i c_i)$, $t_p = 2n_o c_o/(n_i c_o + n_o c_i)$, $c_o = \boldsymbol{d}_o \cdot \boldsymbol{n}$, $c_i = \boldsymbol{d}_i \cdot \boldsymbol{n}$. This is the library operator `fresnel_transmission_matrix`; it is applied pointwise with the true normal of the oval. No propagation has happened yet: $\boldsymbol{E}_t(Q)$ is the transmitted field on Σ.

3 From the field on Σ to the focal angular spectrum

We write the image-space field as a superposition of plane waves,

$$\boldsymbol{E}(\boldsymbol{r}) = \int_{|\boldsymbol{\kappa}| \le k_0 \mathrm{NA}} \boldsymbol{A}(\boldsymbol{\kappa})\, e^{i[\boldsymbol{\kappa}\cdot\boldsymbol{\rho} + k_z(z - z_i)]}\, d^2\kappa, \qquad k_z = \sqrt{k_i^2 - \kappa^2},\ k_i = n_i k_0, \tag{4}$$

with $\boldsymbol{\rho}$ measured from the axis. The task is to obtain $\boldsymbol{A}(\boldsymbol{\kappa})$ from $\boldsymbol{E}_t$ on Σ.

3.1 Exact statement

The field just behind Σ has the local form $\boldsymbol{E}(\boldsymbol{x}) \simeq \boldsymbol{E}_t(Q)\, e^{ik_0\Phi(\boldsymbol{x})}$, where Φ is the optical path from P_0. Its angular spectrum is an integral over Σ (for example the Stratton–Chu or Kirchhoff representation followed by a Fourier transform, or equivalently our operator Q^{-1} followed by R):

$$\boldsymbol{A}(\boldsymbol{\kappa}) = \int_\Sigma \mathsf{K}(\boldsymbol{\kappa}, Q)\, \boldsymbol{E}_t(Q)\, e^{\,i[k_0\Phi(Q) - \boldsymbol{\kappa}\cdot\boldsymbol{\rho}_Q - k_z(z_Q - z_i)]}\, dA_Q, \tag{5}$$

with a slowly varying kernel K. Over the 4.6 mm aperture the phase spans thousands of wavelengths, so (5) is dominated by its stationary points.

3.2 Stationary phase

The phase in (5) is stationary on Σ where the transmitted wavevector $k_i\boldsymbol{d}_i$ has transverse part $\boldsymbol{\kappa}$. Because the diopter is stigmatic, the transmitted wavefront is a sphere centred on F and there is exactly one such point $Q(\boldsymbol{\kappa})$ for each $\boldsymbol{\kappa}$ in the NA disk. Along each ray of a converging spherical wave the amplitude falls as $1/(\text{distance to } F)$, so the field near Σ is

$$\boldsymbol{E} \simeq \frac{\boldsymbol{a}(\boldsymbol{d}_i)}{L}\, e^{-ik_i L}\, e^{ik_0\Phi_F}, \qquad \boldsymbol{a}(\boldsymbol{d}_i) = L_i\, \boldsymbol{E}_t\big(Q(\boldsymbol{d}_i)\big), \tag{6}$$

with L the distance to F and Φ_F the constant optical path to the focus. This is the Debye setting: a spherical wave of angular amplitude $\boldsymbol{a}$ converging on F. Its field near F is

$$\boldsymbol{E}(\boldsymbol{r}) = -\frac{ik_i}{2\pi} \int \boldsymbol{a}(\boldsymbol{s})\, e^{ik_i \boldsymbol{s}\cdot(\boldsymbol{r} - F)}\, d\Omega, \qquad d\Omega = \frac{d^2\kappa}{k_i^2 \cos\theta}, \tag{7}$$

and comparison with (4) gives, up to the constant phase $-i$ common to all components,

$$\boxed{\boldsymbol{A}(\boldsymbol{\kappa}) = \frac{L_i}{2\pi\, k_i \cos\theta}\, \boldsymbol{E}_t\big(Q(\boldsymbol{\kappa})\big)} \tag{8}$$

The Hessian of the phase in (5) at the stationary point produces the same factor $L_i/\cos\theta$; the two derivations agree.

3.3 Energy conservation

Map (8) conserves power patch by patch. The power leaving dA is $\frac{n_i}{2Z_0}|\boldsymbol{E}_t|^2(\boldsymbol{d}_i\cdot\boldsymbol{n})\,dA$. That patch subtends $d\Omega=(\boldsymbol{d}_i\cdot\boldsymbol{n})\,dA/L_i^2$ at F, and $d^2\kappa=k_i^2\cos\theta\,d\Omega$. The spectral power of the same patch is

$$\tfrac{n_i}{2Z_0}(2\pi)^2|\boldsymbol{A}|^2\cos\theta\,d^2\kappa=\tfrac{n_i}{2Z_0}(2\pi)^2\frac{L_i^2|\boldsymbol{E}_t|^2}{(2\pi k_i\cos\theta)^2}\cos\theta\,k_i^2\cos\theta\,d\Omega=\tfrac{n_i}{2Z_0}|\boldsymbol{E}_t|^2(\boldsymbol{d}_i\cdot\boldsymbol{n})\,dA.$$

The notebook checks the integrated balance to better than 10^{-9}.

3.4 What is neglected

Stationary phase keeps the ray contribution of each surface point and drops

- diffraction from the aperture rim, of relative size $\sim N_F^{-1/2}$ with Fresnel number $N_F=n_ia^2/(\lambda_0 z_i)\approx 2.4\times10^3$;
- curvature corrections of order $1/(k_iL_i)\sim10^{-5}$;
- multiple reflections and any field not launched from P_0.

It is therefore the asymptotic limit of Q^{-1} followed by the exact angular spectrum, not the discrete operator itself. The lattice operators $Q\to T\to Q^{-1}\to R$ would require about 10^4 samples per axis across the aperture and are not used here. A direct test would compare both routes on a scaled-down diopter while N_F grows.

4 Azimuthal structure and transfer weights

For the entrance field (2), rotational symmetry of Σ fixes the dependence on the spectral azimuth ψ ($\boldsymbol{\kappa}=\kappa(\cos\psi,\sin\psi)$; the corresponding surface point has $\phi=\psi+\pi$):

$$A_x=c_0+c_2\cos2\psi,\qquad A_y=c_2\sin2\psi,\qquad A_z=c_z\cos\psi,\tag{9}$$

with c_0,c_2,c_z functions of κ. The notebook confirms this form at an independent azimuth to 10^{-12}.

The two channels are characterised by the transfer weights

$$w_p(r)=A_x\big|_{a_p=1,\,a_s=0,\,\psi=0},\qquad w_s(r)=A_x\big|_{a_p=0,\,a_s=1,\,\psi=\pi/2}.\tag{10}$$

Through (2), (3) and (8) they contain the Fresnel coefficients t_p,t_s, the entrance obliquity, the tilt of $\boldsymbol{p}_i$ out of the transverse plane, and the ray-tube factor $L_i/\cos\theta$. For general a_p,a_s the transverse spectrum is

$$\boldsymbol{A}_\perp=a_pw_p\cos\psi\,\hat{\boldsymbol{\kappa}}-a_sw_s\sin\psi\,\hat{\boldsymbol{\psi}},\tag{11}$$

so $c_0=\frac12(a_pw_p+a_sw_s)$ and $c_2=\frac12(a_pw_p-a_sw_s)$. Both weights fall towards the rim, and w_s falls faster; with $a_p=a_s=1$ this makes $c_2\neq0$ and c_0 non-uniform.

5 Compensation

Choose

$$a_p=C\,\frac{w_p(0)}{w_p(r)},\qquad a_s=C\,\frac{w_s(0)}{w_s(r)}.\tag{12}$$

Then $a_pw_p=a_sw_s=C\,w_p(0)$ for every r ($w_p(0)=w_s(0)$ on axis), so $c_2=0$ and $\boldsymbol{A}_\perp=C\,w_p(0)\,\boldsymbol{e}_x$ is uniform over the NA disk. The constant C equalises the incident power through Σ, $P_{\text{in}}=\frac{n_o}{2Z_0}\int_\Sigma|\boldsymbol{E}_0|^2(\boldsymbol{d}_o\cdot\boldsymbol{n})\,dA$, with that of the uncompensated field. The longitudinal part A_z follows from transversality and is not prescribed.

6 Focal field

In the focal plane $z = z_i$ the ψ integrals of (4) with (9) follow from $\int_0^{2\pi} \cos(m\psi)\, e^{ix\cos(\psi-\varphi)} d\psi = 2\pi i^m J_m(x) \cos m\varphi$:

$$E_x = F_0 - F_2 \cos 2\varphi, \qquad E_y = -F_2 \sin 2\varphi, \qquad E_z = iF_z \cos\varphi, \tag{13}$$

$$F_0 = 2\pi \int_0^{k_0 \mathrm{NA}} c_0 J_0(\kappa\rho)\, \kappa\, d\kappa, \quad F_2 = 2\pi \int_0^{k_0 \mathrm{NA}} c_2 J_2(\kappa\rho)\, \kappa\, d\kappa, \quad F_z = 2\pi \int_0^{k_0 \mathrm{NA}} c_z J_1(\kappa\rho)\, \kappa\, d\kappa. \tag{14}$$

The integrals are evaluated with Gauss–Legendre quadrature (400 nodes; doubling changes them by less than 10^{-10} relative).

For the compensated field $c_2 = 0$ and c_0 is constant, so

$$\boldsymbol{E}_\perp = 2\pi c_0\, (k_0 \mathrm{NA})^2\, \frac{J_1(v)}{v}\, \boldsymbol{e}_x, \qquad v = k_0\, \mathrm{NA}\, \rho, \tag{15}$$

the Airy amplitude, with first zero at $r_A = 3.8317/(k_0 \mathrm{NA}) = 0.61\, \lambda_0/\mathrm{NA}$.

7 Results of notebook 04

Quantity	Uncompensated	Compensated
First transverse minimum along x (μm)	0.5177	0.3858
First transverse minimum along y (μm)	0.5471	0.3858
Airy first-zero radius r_A (μm)	0.38579	0.38579
Departure of $\boldsymbol{E}_\perp$ from the Airy focus	47.2 %	$< 10^{-4}$ %
$\max \lvert E_x/E_x(0) - 2J_1(v)/v \rvert$	–	4×10^{-14}
Transmitted / incident power at Σ	0.858	0.537

The compensation needs entrance amplitudes of 8–9 times the axial value near the rim, where incidence on Σ approaches grazing (88.6°) and both transfer weights fall to about 5 % of their axial value. This is why the recovered Airy focus costs throughput at equal incident power.